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# The central limit theorem

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Probability and Statistics for Engineers Fall 2026

Monday, November 16

Midterm 2

Wednesday, November 18

Sums, averages, and standardization

Friday, November 20

Approximation and interpretation

# Midterm 2

## Monday, November 16

55 minutes of examination work. Use the separate examination and formula sheet.

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|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope    | Required material taught through November 9, with earlier foundations.                                                                                        |
| Core     | Expectation, variance, joint and conditional laws, covariance, elementary CDF transformations, Markov and Chebyshev bounds, and the finite-variance weak LLN. |
| Excluded | CLT, inference, and stochastic processes.                                                                                                                     |

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No new lecture material in this meeting.

## A sum and an average

Let  $X_1, \dots, X_n$  be iid with mean  $\mu$  and variance  $\sigma^2$ .

$$S_n = X_1 + \dots + X_n, \quad \bar{X}_n = \frac{S_n}{n}.$$

If each observation measures time in seconds, what does each quantity measure?

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Predict the scales first. Then derive them.

## Mean and variance of a sum

Assume  $\mathbb{E}[X_i] = \mu$  and  $\text{Var}(X_i) = \sigma^2 < \infty$  for iid observations. Derive the moments of  $S_n$  and  $\bar{X}_n$ . Identify where independence enters.

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The variance of an average contains a factor  $1/n^2$  before simplification.

## One standardized quantity

Suppose  $0 < \sigma^2 < \infty$ . Center and divide by the standard deviation:

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}.$$

Express  $Z_n$  using  $\bar{X}_n$ , then derive its mean and variance.

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# The central limit theorem

Let  $X_1, X_2, \dots$  be iid, with

$$\mathbb{E}[X_i] = \mu, \quad 0 < \sigma^2 = \text{Var}(X_i) < \infty.$$

For every fixed real  $z$ ,

$$\lim_{n \rightarrow \infty} P\left(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq z\right) = \Phi(z).$$

Here  $\Phi(z) = P(G \leq z)$  for  $G \sim N(0, 1)$ .

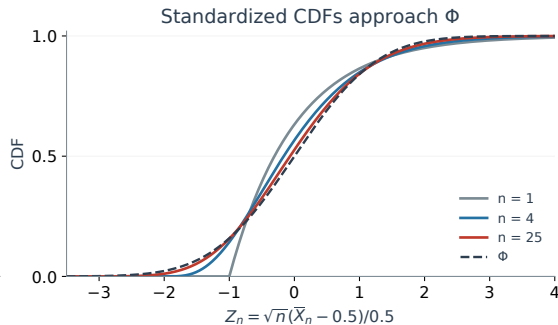
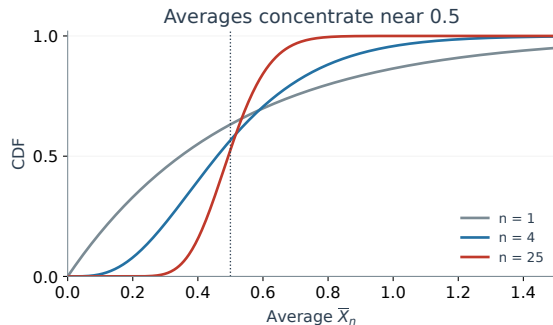
The standardized CDF approaches the standard normal CDF.

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Theorem stated without proof. Harchol-Balter, §9.4; DeGroot–Schervish, §6.3.

# Concentration and standardized fluctuations

$X_i$  are iid exponential with rate 2, so  $\mu = \sigma = 1/2$ .



LLN:  $\bar{X}_n$  concentrates near  $\mu$ . CLT: the CDF of  $Z_n$  approaches  $\Phi$ .

## Exponential measurements

Model independent component times by  $X_i \stackrel{\text{iid}}{\sim} \text{Exp}(2)$ , with rate 2 per second. Recall

$$f_X(x) = 2e^{-2x} \ (x > 0), \quad \mu = \frac{1}{2}, \quad \sigma^2 = \frac{1}{4}.$$

For  $n = 100$ , approximate

$$P(0.45 < \bar{X}_n < 0.55).$$

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## Probability for an average

$X_i \stackrel{\text{iid}}{\sim} \text{Exp}(2)$ ,  $n = 100$ ,  $\mu = \sigma = 1/2$ . Find the sampling SD and translate  $0.45 < \bar{X}_n < 0.55$  to the  $Z_n$  scale.

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$\Phi(1) \approx 0.841345$ . The normal replacement is the only approximate step.

## The same event for the sum

With  $n = 100$ ,  $S_{100} = 100\bar{X}_{100}$ .

$$\{0.45 < \bar{X}_{100} < 0.55\} = \{45 < S_{100} < 55\}.$$

| Quantity        | Mean | Standard deviation |
|-----------------|------|--------------------|
| $\bar{X}_{100}$ | 0.5  | 0.05               |
| $S_{100}$       | 50   | 5                  |

Check the standardized endpoints using the sum.

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## Four times as many observations

Keep  $X_i \stackrel{\text{iid}}{\sim} \text{Exp}(2)$  and the same tolerance. For  $n = 400$ , approximate  $P(0.45 < \bar{X}_n < 0.55)$ . Explain the change from  $n = 100$  before calculating.

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$$\Phi(2) \approx 0.977250.$$

## Equality, bound, limit, approximation

Assume iid observations with mean  $\mu$  and  $0 < \sigma^2 < \infty$ . Identify the status of each statement.

| Statement                                                              | Status |
|------------------------------------------------------------------------|--------|
| $\text{Var}(\bar{X}_n) = \sigma^2/n$                                   |        |
| $P( \bar{X}_n - \mu  \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2)$ |        |
| $P(Z_n \leq z) \rightarrow \Phi(z)$ for fixed $z$                      |        |
| $P(Z_{100} \leq 1) \approx \Phi(1)$                                    |        |

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## Center and scale

iid observations have mean  $\mu$  and variance  $\sigma^2 > 0$ . Complete the denominators:

$$\frac{S_n - n\mu}{\quad}, \quad \frac{\bar{X}_n - \mu}{\quad}.$$

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## Variance of a Bernoulli variable

Let  $I$  equal 1 with probability  $p$  and 0 with probability  $1 - p$ . Use  $I^2 = I$  to derive its variance.

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An indicator counts one success. Its square has the same value.

## Mean and variance of a count

Let  $I_1, \dots, I_n$  be iid Bernoulli( $p$ ), with  $0 < p < 1$ , and

$$K_n = \sum_{i=1}^n I_i \sim \text{Bin}(n, p).$$

Derive the moments and the standardized count.

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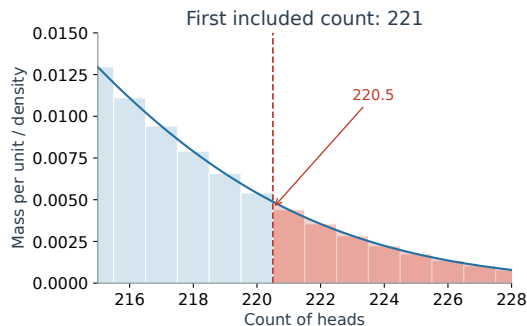
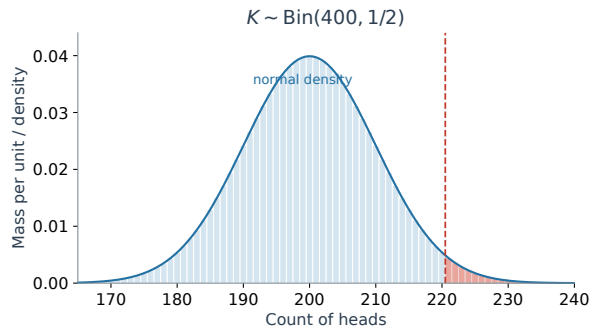
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Check the trial assumptions before using a binomial model.

# Counts occupy cells

For integer  $k$ ,  $\{K > k\} = \{K \geq k + 1\}$ . The first included cell starts at  $k + 1/2$ .



$$\begin{array}{ccc} K \leq k & K \geq k & a \leq K \leq b \\ \text{up to } k + 1/2 & \text{from } k - 1/2 & \text{from } a - 1/2 \text{ to } b + 1/2 \end{array}$$

Each unit-width bar has area equal to its probability mass.

## More than 220 heads

A fair coin is tossed independently 400 times. Let  $K$  count the heads.

$$K \sim \text{Bin}(400, 1/2).$$

Approximate  $P(K > 220)$ .

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Adapted from the supplied STAT131 Discussion 10.

## Normal approximation to the tail

$K \sim \text{Bin}(400, 1/2)$  has mean 200 and SD 10. Use the boundary 220.5 for  $\{K > 220\}$ .

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| Method                | Probability | Absolute error |
|-----------------------|-------------|----------------|
| Exact sum (rounded)   | 0.0201154   | —              |
| Normal, no correction | 0.0227501   | 0.0026348      |
| Normal, corrected     | 0.0201822   | 0.0000668      |

Exact benchmark:  $2^{-400} \sum_{j=221}^{400} \binom{400}{j} \cdot \Phi(2.05) \approx 0.979818$ .

## A large sample can still give a poor approximation

Let  $K \sim \text{Bin}(1000, 0.001)$ . Then  $\mathbb{E}[K] = 1$  and  $\text{Var}(K) = 0.999$ .

$$P(K \geq 1) = 1 - (0.999)^{1000} \approx 0.632305.$$

Corrected normal approximation:  $1 - \Phi\left(\frac{0.5 - 1}{\sqrt{0.999}}\right) \approx 0.691551.$

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The sample size alone cannot certify accuracy.

## Planning a sample size

lid rod lengths have mean 50 mm and known SD 8 mm. Choose  $n$  for a target probability 0.95 of  $|\bar{X}_n - 50| < 2$  mm.

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62 is an approximate planning choice. 320 is sufficient from the variance alone.

## A count threshold

Independent tests each pass with probability  $1/2$ . Among 100 tests, let  $K$  be the number that pass. Approximate  $P(K \geq 60)$ . State the model, identify the first included cell boundary, and interpret your answer.

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$$\Phi(1.9) \approx 0.971283.$$

## What the approximation establishes

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|               |                                                                      |
|---------------|----------------------------------------------------------------------|
| Assumptions   | iid observations with positive finite variance.                      |
| Quantity      | Centered sum or average, divided by its own SD.                      |
| Finite sample | A normal probability approximation whose accuracy needs judgment.    |
| Sampling SD   | $\sigma/\sqrt{n}$ for $\bar{X}_n$ , often called its standard error. |

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Next lecture: use sampling variability to construct an interval for a mean.

## Optional: independent signal noise

$X_i \stackrel{\text{iid}}{\sim} \text{Unif}(-1, 1)$  models additive component noise. For  $n = 100$ , approximate  $P(|S_n| < 10)$  and compare with Chebyshev.

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Harchol-Balter, Example 9.9, pp.177–178.  $\Phi(\sqrt{3}) \approx 0.958368$ .

## Optional: a smaller tail

Let  $K \sim \text{Bin}(900, 1/2)$ . Approximate  $P(K > 495)$ . Compare the direct normal calculation with the continuity-corrected result.

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DeGroot–Schervish, Example 6.3.2 and §6.4. Supplied Lec10, frames 21–22.

## Optional: one count value

For  $K \sim \text{Bin}(20, 1/2)$ , approximate  $P(K = 10)$ . Which interval represents the mass at ten?

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DeGroot–Schervish, Example 6.4.4.

## Optional: repeated measurements

Let  $X_i$  be iid unbiased measurements of a true value  $\theta$ , with SD  $\sigma > 0$ . For  $n = 25$ , estimate  $P(|\bar{X}_n - \theta| < \sigma/4)$ . Give a Chebyshev lower bound as well.

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Supplied STAT131 problems, §4.2, Q1.  $\Phi(1.25) \approx 0.894350$ .

## Optional: a 99 percent precision target

Keep the same measurement model and tolerance  $\sigma/4$ . Find a normal planning size and a Chebyshev sufficient size for a target probability 0.99.

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Supplied STAT131 problems, §4.2, Q2.

## Selected reading

### CLT and standardization

Harchol-Balter, *Introduction to Probability for Computing*, supplied 2024 version, §9.4, pp.176–178.

DeGroot–Schervish, *Probability and Statistics*, 4th ed., §6.3, pp.361–363.

### Continuity correction

DeGroot–Schervish, §6.4, pp.371–374.

### Optional practice

Week 8 optional practice and checked answers. The notes contain the required statements and derivations.

The CLT proof, delta method, and more general limit theorems are outside this week's required material.